



COURSE SYLLABUS						
Course Code	COMP 016	Course Title	WEB DEVELOPMENT		Course Credit	3
Course Description		Web development is the process of creating and maintaining websites. It involves designing, coding, and building web pages using languages like HTML, CSS, and JavaScript to make them functional, user-friendly, and visually appealing.				
Pre-Requisites		Comp 009 – Object Oriented Programming	Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				
INSTITUTIONAL LEARNING OUTCOMES (ILO)		As a polytechnic state university, PUP shall develop its students to possess: Critical and Creative Thinking. Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world. Effective Communication. Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances. Strong Service Orientation. Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors. Adept and Responsible Use or Development of Technology. Graduates demonstrate optimized and responsible use of state-of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills. Passion for Lifelong Learning. Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.				



	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10. Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11. Produce technologies for commercialization of livelihood improvement.

PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
	Apply knowledge of computing, science, and mathematics appropriate to the discipline.									
	Analyze complex problems, and identify and define the computing requirements appropriate to its solution									
	Identify and analyze user needs and take them into account in the selection, creation, evaluation									



	and administration of computer-based systems.									
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.									

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	Integrate IT-based solutions into the user environment effectively.												
	Assist in the creation of an effective IT project plan.												
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession												
	Understand best practices and standards and their applications.												
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.												
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.												
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.												
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.												
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development												

COURSE LEARNING OUTCOMES (CLO)		Alignment to PLOs							
		1	2	3	4	5	6	7	8
	Explain the fundamental concepts, principles, and components of web development, including the client-server model and the role of front-end technologies.								
	Construct well-structured web pages using proper HTML syntax, tags, and semantic elements following W3C standards.								
	Apply CSS styling techniques to enhance the visual layout, responsiveness, and accessibility of web pages according to modern UI/UX principles.								



	Develop interactive and dynamic web content using JavaScript to improve user engagement and interface behavior.								
	Integrate HTML, CSS, and JavaScript into cohesive and functional web applications that follow usability and design best practices.								
	Utilize frameworks and libraries (e.g., Bootstrap) to create responsive, mobile-friendly, and scalable web interfaces.								
	Demonstrate proficiency in deploying and maintaining websites using online hosting platforms, version control systems (e.g., GitHub), and domain management tools.								
	Collaborate effectively in team-based projects to plan, design, develop, and present a complete web application demonstrating creativity, problem-solving, and technical competence.								



OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/Topics	Instructional Delivery Design		Assessment		Alignm ent to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)			1	2	3	4
					Face-to-Face	FLTAs				
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP - Gain an overview of the course subject and requirements	- Orientation about PUP Vision, Mission, and 10 Pillars - Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.		Formative Assessment (Sharing of Learning)		X			
2	- Understand the fundamentals and scope of web development - Identify and use HTML structure and tags correctly	- Course orientation, Introduction to web development, Internet & WWW overview - HTML syntax, elements, attributes, and structure	Discuss the purpose of web development and course overview and Hands-on demo: creating a basic webpage	Recorded orientation video, interactive discussion forum, online quiz about web concepts.	Attendance and participation check.	Forum participation and short quiz.				
3	Apply HTML formatting and multimedia elements	Text formatting tags, hyperlinks, images, audio, and video embedding	Demonstration using HTML tags for media; group task to add media to a webpage.	Step-by-step screencast lesson; peer review of	Lab Activity 2 on creating a simple or static website.	Peer-reviewed HTML project.				



				uploaded HTML pages.					
4	Develop structured web pages using lists, tables, and forms	Ordered/unordered lists, tables, input forms, and basic form validation	Hands-on coding session in lab; instructor provides feedback.	Virtual coding workshop using a shared IDE; discussion forum for form design ideas.	Lab Activity 3.	Interactive quiz on HTML forms.			
5	Apply CSS syntax to style web pages	Introduction to CSS, syntax, selectors, colors, fonts, and box model	Instructor-led demo on styling; students apply CSS to a sample HTML file.	Recorded CSS tutorial and styling challenge on a provided HTML template.	Quiz #1 – HTML & CSS Basics. (Covers basic HTML tags and CSS syntax.)	Styling Activity Submission.			
6	Create layouts using advanced CSS features	Flexbox, grid, and responsive design techniques	Workshop where students build responsive layouts on desktop.	Live coding session with breakout groups for layout debugging.	Lab Activity 4 – Responsive Layout. (Hands-on coding exercises.)	Layout Design Submission.			
7	Integrate external CSS and frameworks	Linking CSS files, using Bootstrap for responsive design	Group coding using Bootstrap; instructor-guided practice.	Online workshop with downloadable Bootstrap template.	Quiz #2 – CSS and Layouts (Checks understanding of Flexbox and Grid.)	Bootstrap webpage submission.			



8	Implement JavaScript for basic interactivity Handle user input using JavaScript	JS syntax, variables, functions, and DOM basics Event listeners, form validation, and alert messages	Hands-on session coding form validation scripts; real-time instructor support.	Virtual session with problem-solving activity on JS events.	Quiz #3 – JavaScript Basics. (Focuses on syntax, events, and functions.)	JS validation assignment.				
9	MIDTERM WRITTEN EXAMINATION									
10	Utilize advanced JavaScript features	Arrays, loops, conditionals, ES6 features	Lecture-coding integration; students create dynamic data display.	Interactive coding challenges using online IDE with instructor feedback.	Coding Challenge. (Hands-on coding exercises.)	Online Quiz (JS Arrays & Loops).				
11	Apply DOM manipulation for dynamic webpages	DOM structure, traversing, event handling	Instructor demonstrates DOM scripting; students modify an existing site.	DOM coding task in JSBin with auto-check.	Lab Activity 6 – DOM Project. (Hands-on coding exercises.)	Mini DOM project submission.				
12	Combine HTML, CSS, and JS into integrated webpages	Project integration and interface testing	Group work in lab; project consultation and debugging session.	Virtual consultation with feedback via Google Meet or LMS.	Midterm Project Submission. (Integration of HTML, CSS, and JS into a single webpage.)	Prototype upload for review.				
13	Understand web hosting and deployment	Web hosting basics, FTP, GitHub Pages, Netlify	Demonstration on deploying websites; students deploy sample page.	Recorded video + step-by-step online hosting guide.	Quiz #4 – Web Deployment. (Tests knowledge of hosting and GitHub Pages.)	Hosting activity submission.				
14	Understand backend	Introduction to servers, HTTP requests, and	Lecture-demo with simple backend setup.	Pre-recorded lecture with	Lab Activity 7 – Backend Basics. (Hands-on coding	Short quiz on				



	development basics	PHP/Node.js overview		coding notes.	exercises.)	backend concepts.				
15	Connect front-end and backend using form data	Handling GET/POST methods, basic database concept	Guided code-along connecting forms to backend.	Live simulation using online backend tool (e.g., Replit).	Data form handling activity.	Online code submission.				
16	Work collaboratively on a web project	Project proposal, wireframing, and team collaboration	Group consultation and feedback with instructor.	Collaborative online meeting for proposal presentation.	Group Proposal Presentation.	Proposal video upload.				
17	Develop and finalize a complete web application Present and evaluate the final web project	Project implementation, integration, and debugging Project demonstration, reflection, and evaluation	Supervised coding session and final review. Live final presentation and oral defense.	Group collaboration on GitHub and Google Meet check-ins. Recorded presentation with instructor evaluation.	Project progress review. Final Project Defense.	Peer feedback submission. Final Project Submission and Video Defense.				
18	FINAL WRITTEN EXAMINATION (It depends on the professor whether the final examination will be conducted in written or laboratory form)									



		Resources and Applying Strategies in Teaching Prose Competency 4: Recognizing Assessment in Teaching Prose	achievement which are consistent with the selected competencies • Learning Plan Making according to the English curricula that develops higher order thinking skills of learners through the use of prose and dramatic texts		Quizzes					
18	FINAL EXAMINATION									
LIST OF REFERENCES BOOKS: WEBSITE: Bootstrap. (n.d.). <i>Bootstrap documentation</i>. https://getbootstrap.com/docs/ Codecademy. (n.d.). <i>Learn web development</i>. https://www.codecademy.com/learn/paths/web-development Coursera. (n.d.). <i>Web development courses</i>. https://www.coursera.org/browse/computer-science/mobile-and-web-development CSS-Tricks. (n.d.). <i>CSS-Tricks: Tips, tricks, and techniques on using Cascading Style Sheets</i>. https://css-tricks.com freeCodeCamp. (n.d.). <i>Learn to code — Build projects — Earn certifications</i>. https://www.freecodecamp.org GeeksforGeeks. (n.d.). <i>Web development tutorials</i>. https://www.geeksforgeeks.org/web-development/ GitHub. (n.d.). <i>GitHub guides</i>. https://guides.github.com Google Developers. (n.d.). <i>Web fundamentals</i>. https://developers.google.com/web MDN Web Docs. (n.d.). <i>Resources for developers, by developers</i>. https://developer.mozilla.org Node.js Foundation. (n.d.). <i>Node.js documentation</i>. https://nodejs.org/en/docs/ Stack Overflow. (n.d.). <i>Where developers learn, share, & build careers</i>. https://stackoverflow.com The Odin Project. (n.d.). <i>Your career in web development starts here</i>. https://www.theodinproject.com TutorialsPoint. (n.d.). <i>Web development tutorials</i>. https://www.tutorialspoint.com/web_development_tutorials.htm W3C – World Wide Web Consortium. (n.d.). <i>Web standards</i>. https://www.w3.org/ W3Schools. (n.d.). <i>Web development tutorials</i>. https://www.w3schools.com										



CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students' Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.
We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.
Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.
Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.
Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.
Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.
Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a



	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.	
COURSE REQUIREMENT/S	GRADING SYSTEM	
- Students are required to create a Web Project Plan, complete hands-on coding activities, and develop both a Midterm Interactive Web Page and a Final Web Application Project. They must also conduct a Web Development Demonstration to present and explain their system’s features and functionality.	Mid Term	Final Term
	Class Standing 70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper	Class Standing 70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper
	Midterm Examinations 30%	Final Examinations (Teaching Demonstration) 30%
	100%	100%
Final Grade = (Midterm Grade + Final Term Grade) / 2		
CLASS INFORMATION		FACULTY INFORMATION
Course, Year and Section: BSIT 3 - 1		Name of Faculty: Orlando Mendoza Lingo
Day and Time: MONDAY – 4:00 to 9:00		Consultation Time:
Room: DOST LABORATORY		Office Tel. No./ Mobile Phone No.
School Year: 2025-2026 Semester: 3RD YEAR (First Semester)		E-Mail Address:



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE VICE PRESIDENT FOR CAMPUSES
TAGUIG CAMPUS
Bachelor of Science in Information Technology



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Teaching
Academic Development
Inclusiveness

★★★★★
Social Responsibility
Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

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